

Examiner reports

Q1.

This question was a good source of marks for most students. In part (a), 95% of students correctly identified the outliers. Although some did not gain further marks, rather than offering an explanation as to the difference in Collins and Donovan's values compared to the others they just stated the reason they had picked them out as the outliers, effectively re-stating the co-ordinate values.

In part (b) students often lost the mark for not indicating the strength of the positive correlation.

Q5.

A reasonable number of candidates obtained the correct values for the mode, median and interquartile range. Some candidates even obtained an answer for the mean within the range permitted, but very few scored the 2 marks for the standard deviation. It was evident that most candidates were unsure as to how to treat grouped data with the result that they probably used incorrect mid-points, but the absence of any detail made this uncertain.

Of even more concern was the number of candidates who failed to use the frequencies, particularly when calculating the standard deviation. Calculating the mean and the standard deviation from a frequency distribution or from a grouped frequency distribution must be bread and butter to candidates entered for this paper. Disappointingly, most candidates apparently had no clue as to what was meant by 'measure of spread'. In fact 'mode', 'median', 'mean', 'bar chart', 'histogram', etc were seen with varying frequencies. It should be noted that the standard responses of 'mean and standard deviation' or 'median and interquartile range' also scored no marks.

Q6.

Most candidates obtained the correct values for the mode, median, interquartile range and range, although some candidates treated the data as continuous. A small minority hedged their bets for the mode by quoting values of 253 and 20, and similar numbers did not state the difference between their correct values for the upper and lower quartiles or merely stated the range to be 227 to 271. Each of these three errors lost a mark. A majority of candidates obtained an answer for the mean within the range permitted, but far fewer scored the 2 marks for the standard deviation. It was evident that too many candidates were unsure as to how to treat grouped data with the result that they probably used incorrect mid-points, but the absence of any detail made this uncertain.

Of even more concern was the significant number of candidates who failed to use the frequencies, particularly when calculating the standard deviation. Calculating the mean and the standard deviation from a frequency distribution or from a grouped frequency distribution should be bread and butter to candidates entered for this paper.

Disappointingly, most candidates apparently had no clue as to what was meant by 'measure of spread'. In fact, the most common response was 'mode' but 'mean', 'median', 'bar chart', 'histogram', 'box and whisker', 'stem and leaf', 'correlation' or even 'CLT' were seen with varying frequencies. It should be noted that the standard responses of 'mean and standard deviation' or 'median and interquartile range' also scored no marks.

However, when interquartile range alone was stated, the usual correct reason related to extreme values and, when just standard deviation was quoted, reference was made to it

using all the data.

Q7.

This question, requiring an entirely descriptive answer, was the least well answered. Even so most candidates realised that the product moment correlation coefficient is not appropriate for non-linear relationships.



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